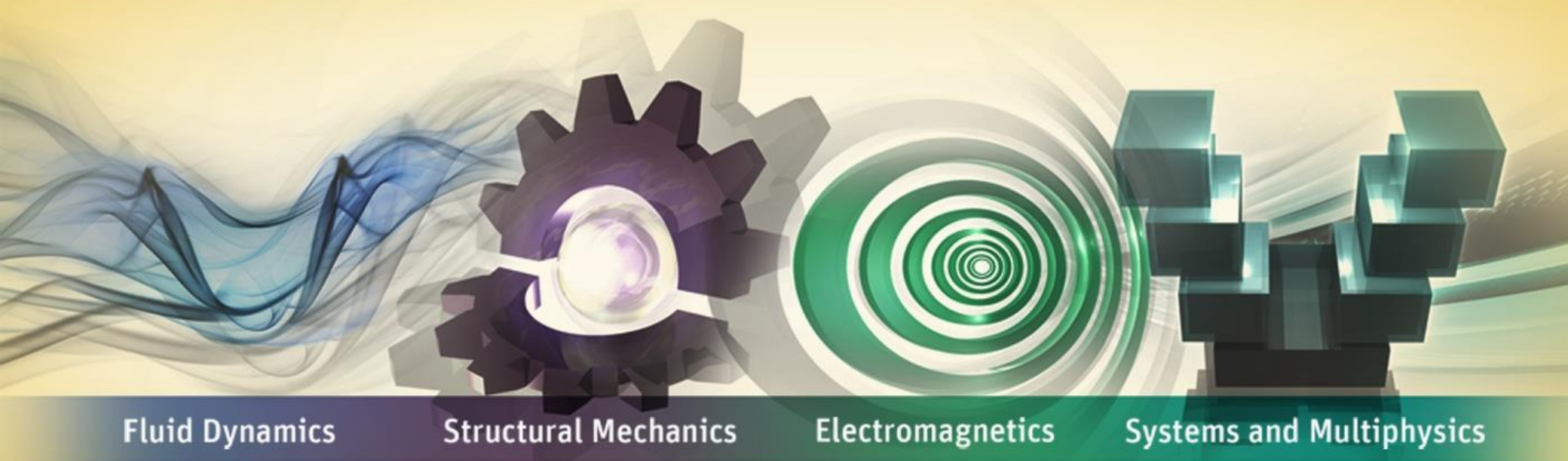


Workshop 2A: Small vs Large Deflection



ANSYS Mechanical Introduction to Structural Nonlinearities

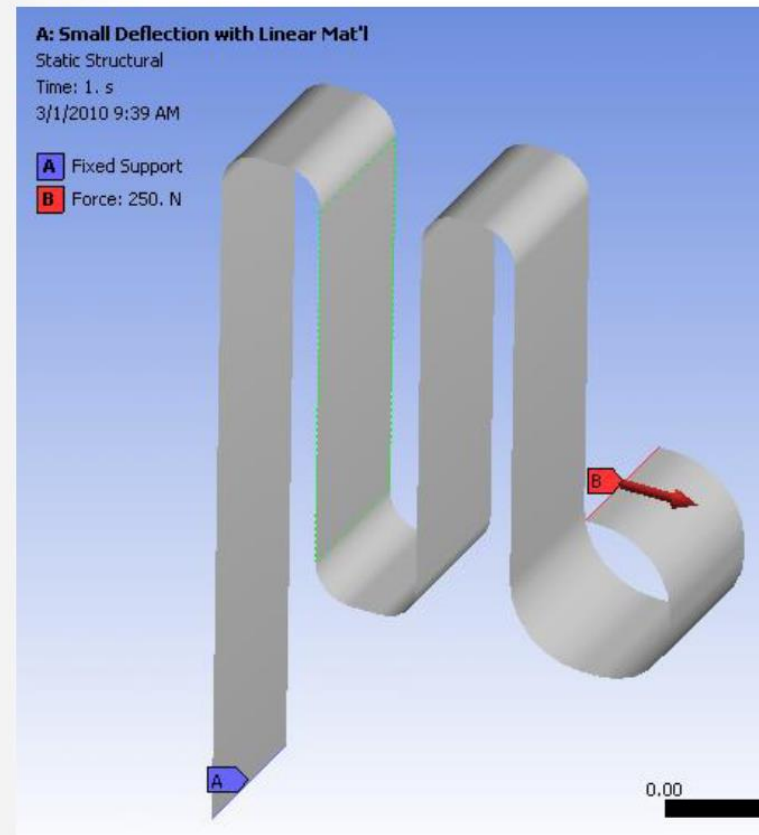
Workshop 2A: Large Deflection

Goal

Compare and contrast results using small deflection theory and large deflection theory on a model with identical loads and boundary conditions.

Model Description

- 3D Spring plate
- Linear steel material
- Meshed with 3D Shell elements
- Fixed support at one end, A
- 250N load at opposite end, B



... Workshop 2A: Large Deflection

Steps to Follow:

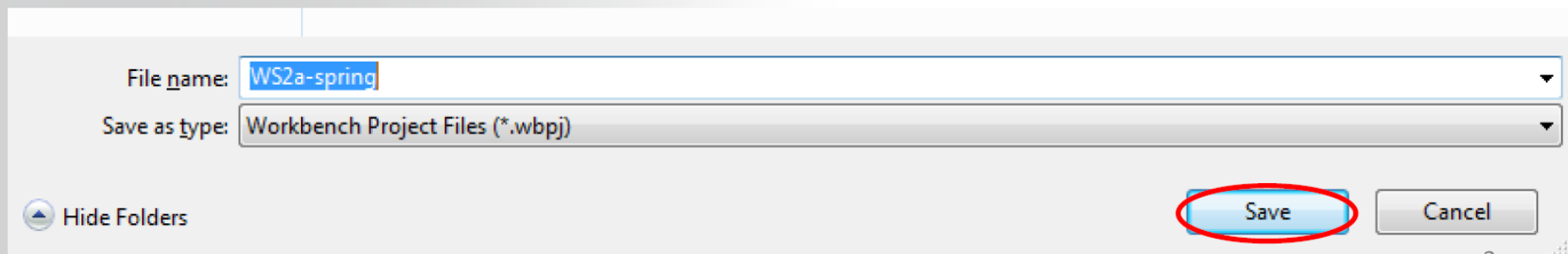
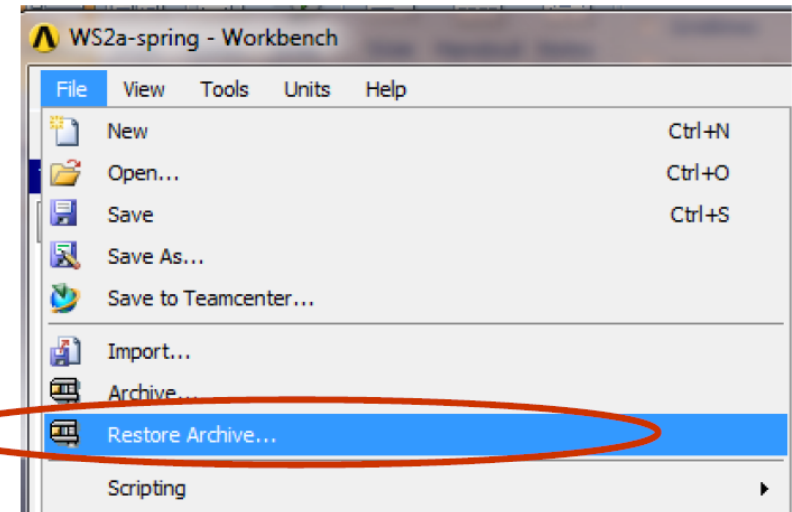
Start an ANSYS Workbench session.

Use File > Restore Archive... browse for existing file “SNL WS2a_spring.wbpz”

- Location of directory provided by instructor

Save as

- File name: “WS2a-spring”
- Save as Type: Workbench Project Files (*.wbpj)

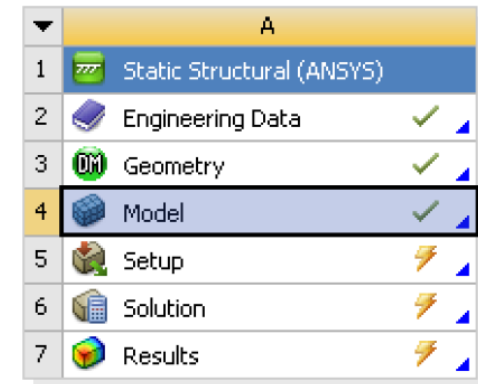


... Workshop 2A: Large Deflection

The Project Schematic should look like the picture to the right.

- From this Schematic, you can see that the Engineering (material) Data and Geometry have already been defined (green check marks).
- It remains to set up and run the FE model in Mechanical
- Open the Engineering Data Cell (highlight and double click OR Right Mouse Button (RMB)>Edit) to verify the linear material properties.
- To see relevant dialog boxes, it might be necessary to go to Utility Menu > View..
 - Click on 'Properties' and 'Outline'
- Verify that the units are in Metric(Tonne,mm,...) system. If not, fix this by clicking on...
 - Utility Menu > Units > Metric(Tonne, mm,...)

Project Schematic



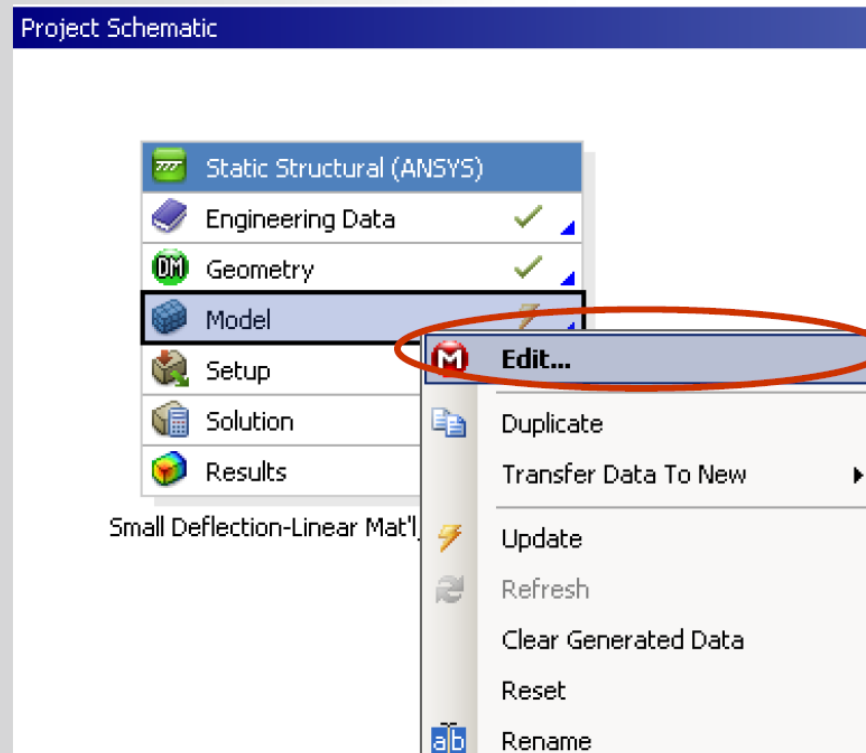
Small Deflection with Linear Mat'l

Properties of Outline Row 3: Structural Steel			
	A	B	C
1	Property	Value	Unit
2	Density	7.85E-09	tonne mm
3	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Young's Modulus and Poisson's Ratio	
8	Young's Modulus	2E+05	MPa
9	Poisson's Ratio	0.3	
10	Bulk Modulus	1.6667E+05	MPa
11	Shear Modulus	76923	MPa
12	Alternating Stress Mean Stress	Tabular	
16	Strain-Life Parameters		
24	Tensile Yield Strength	250	MPa
25	Compressive Yield Strength	250	MPa
26	Tensile Ultimate Strength	460	MPa

... Workshop 2A: Large Deflection

Return to Project Schematic

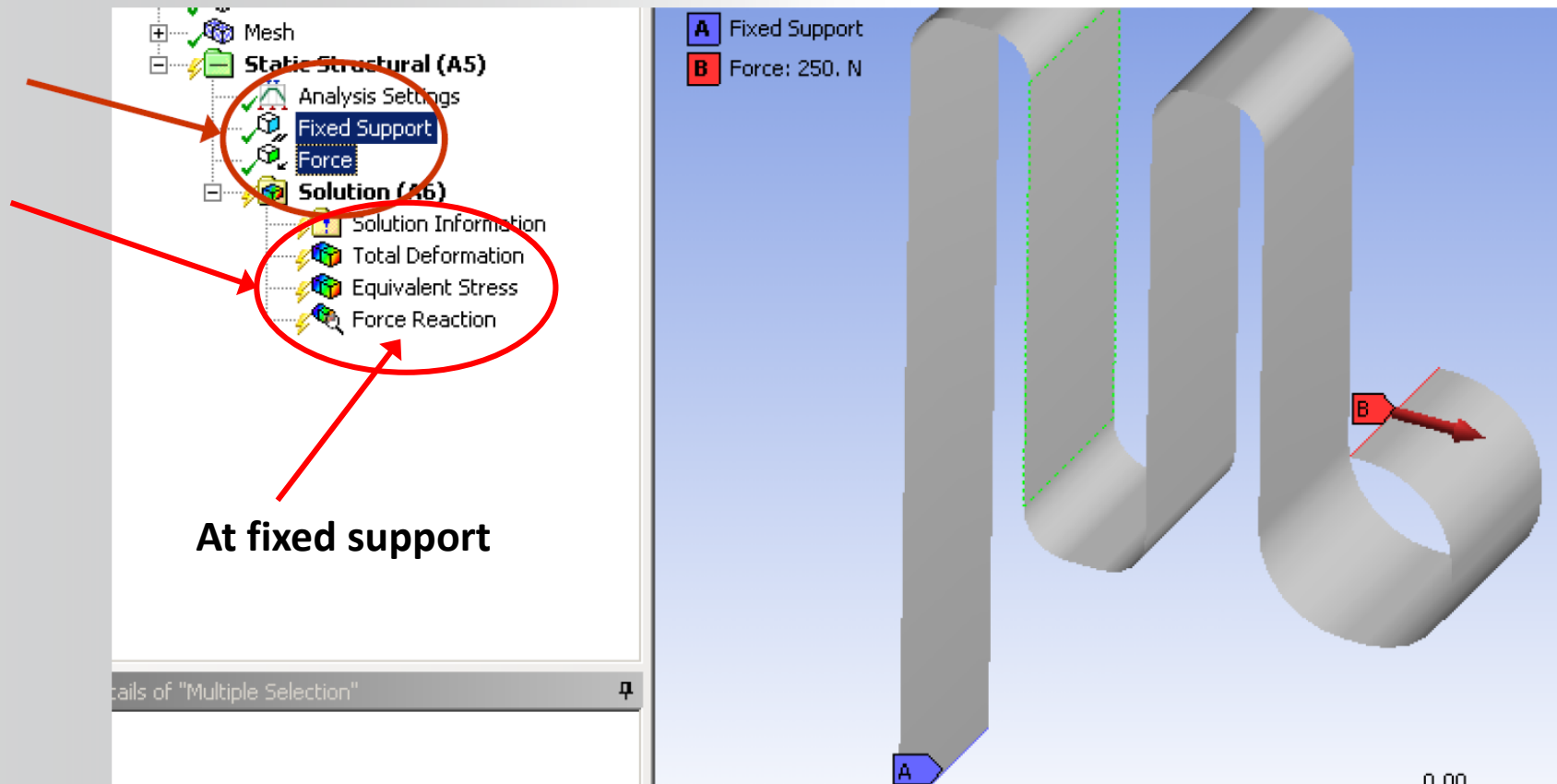
Double click on the Model Cell to open the FE Model (Mechanical Session) (or RMB=>Edit...)



... Workshop 2A: Large Deflection

The spring model is already set up with a fixed boundary condition and a force load on the opposite end.

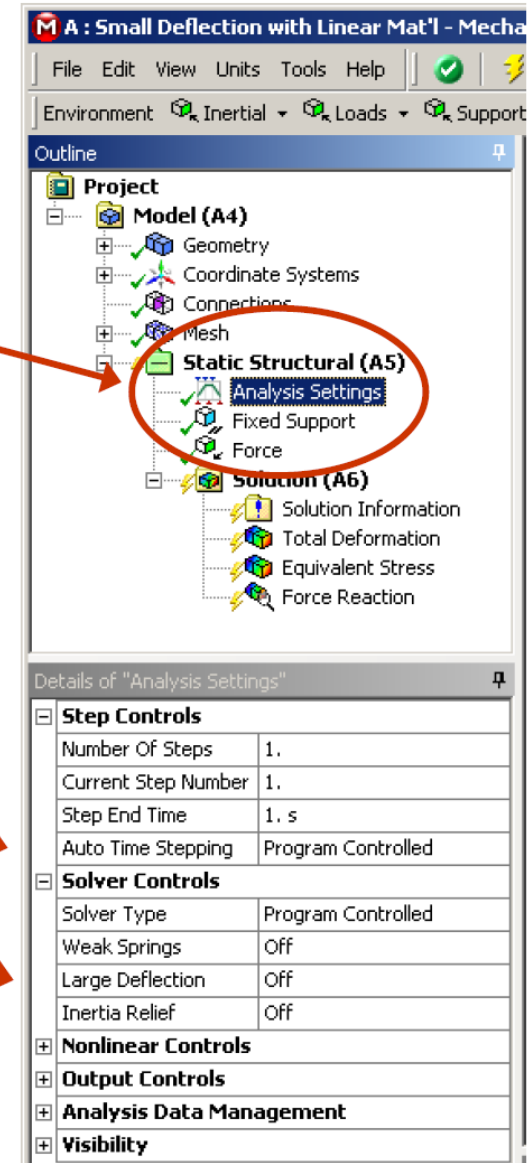
- Highlight the Fixed Support and Force Load to confirm that the model is properly supported, loaded and ready to solve.
- Insert solutions as shown in the picture below.



Note the Analysis Settings Specifications:

- Auto Time Stepping = Program Controlled
- Large Deflection = Off

Run the Solution...



... Workshop 2A: Large Deflection

After solution run is complete, highlight the Solution Information folder and scroll to near the bottom of the output.

- As expected, this solves in one iteration.

The screenshot shows the ANSYS Multiphysics interface. The title bar reads "A : Small Deflection-Linear Mat'l, Static Structural - Mechanical [ANSYS Multiphysics]". The menu bar includes File, Edit, View, Units, Tools, and Help. The toolbar contains various icons for solving and viewing. The Outline pane on the left shows the project hierarchy: Project, Model (A4), Geometry, Coordinate Systems, Mesh, Static Structural (A5), Analysis Settings, Fixed Support, Force, and Solution (A6). The Solution (A6) folder is expanded, showing Solution Information, Total Deformation, Equivalent Stress, and Force Reaction. The Worksheet pane on the right displays the output text. A red circle highlights the following text:

```

*** NODAL LOAD CALCULATION TIMES
TYPE  NUMBER  ENAME      TOTAL CP  AVE CP
1     1182  SHELL181    0.000    0.000000
2         8   SURF156    0.000    0.000000
*** LOAD STEP      1  SUBSTEP      1  COMPLETED.      CUM ITER =      1
*** TIME = 1.00000      TIME INC = 1.00000      NEW TRIANG MATRIX
  
```

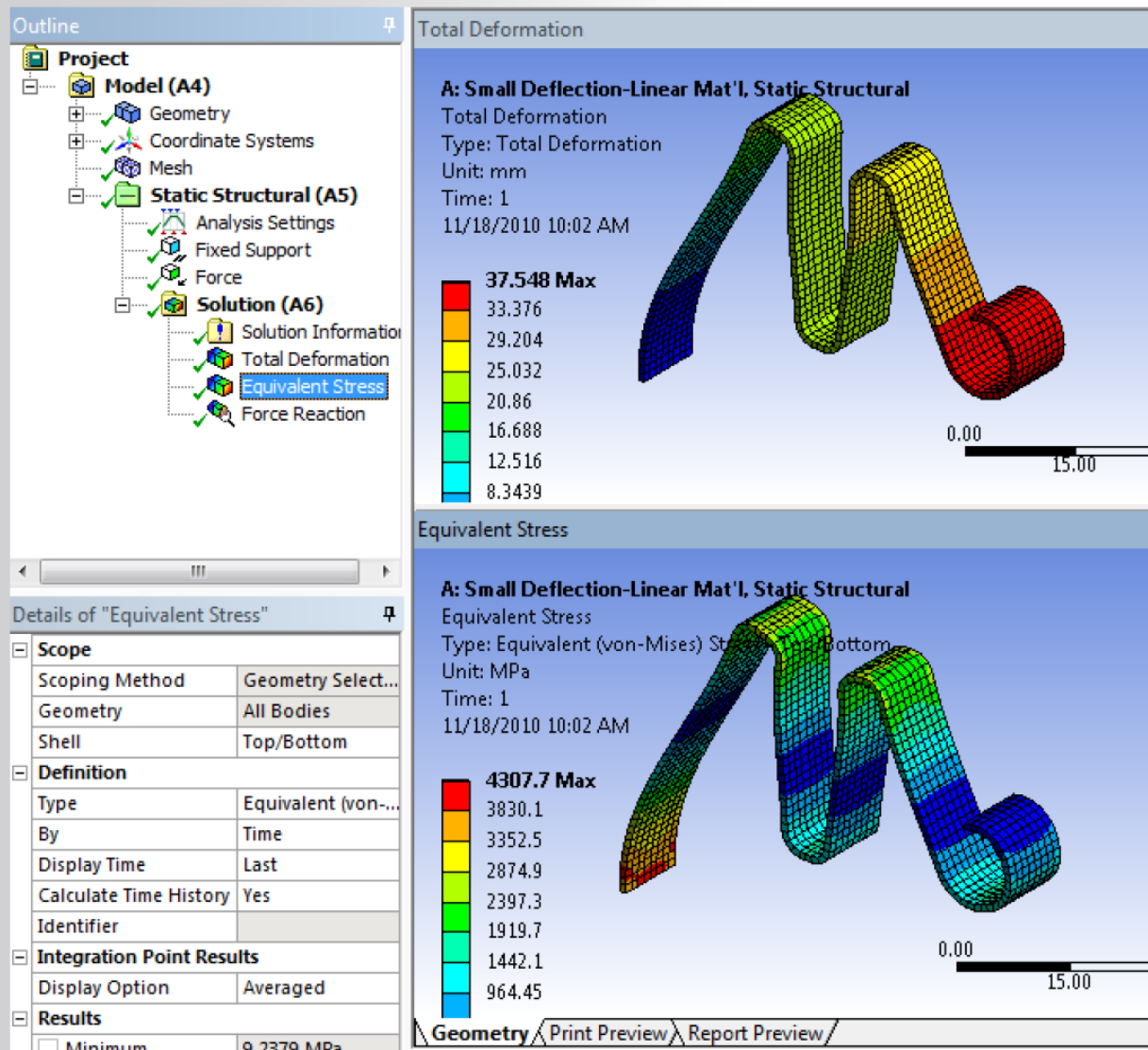
Below this, the ANSYS Binary File Statistics are shown:

```

*** ANSYS BINARY FILE STATISTICS
BUFFER SIZE USED= 16384
0.062 MB WRITTEN ON ELEMENT MATRIX FILE: file.emat
7.000 MB WRITTEN ON ELEMENT SAVED DATA FILE: file.esav
2.000 MB WRITTEN ON ASSEMBLED MATRIX FILE: file.full
3.000 MB WRITTEN ON RESULTS FILE: file.rst
*****
  
```


... Workshop 2A: Large Deflection

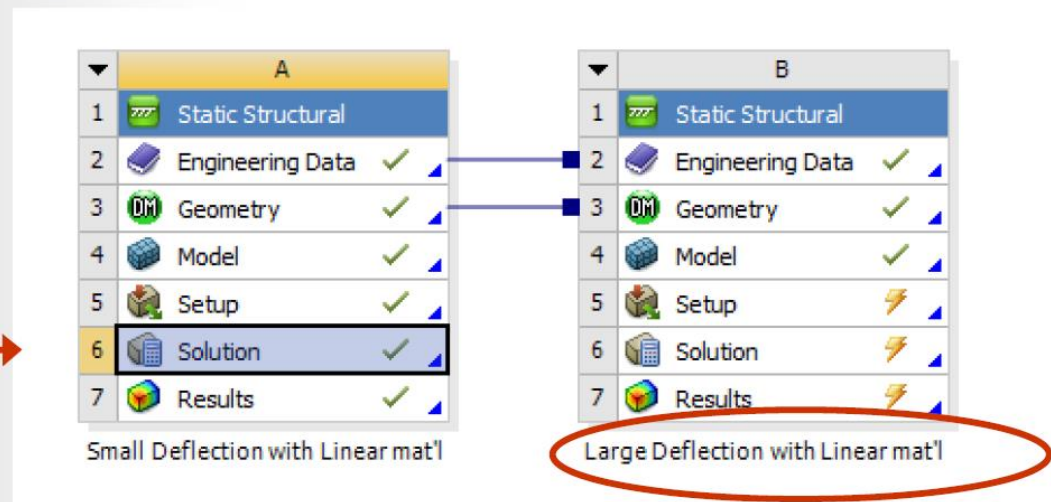
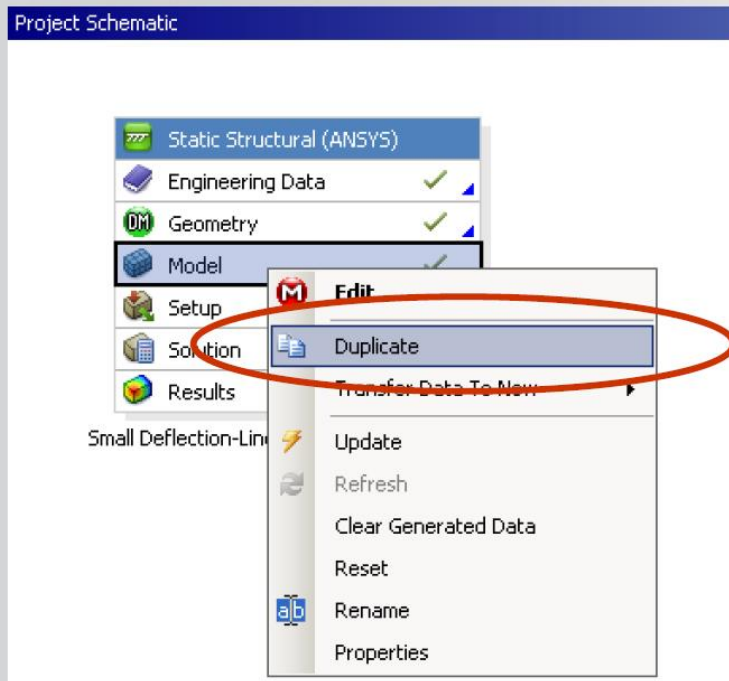
Review the stress and displacement results from this first run.



... Workshop 2A: Large Deflection

Return to the Project Schematic

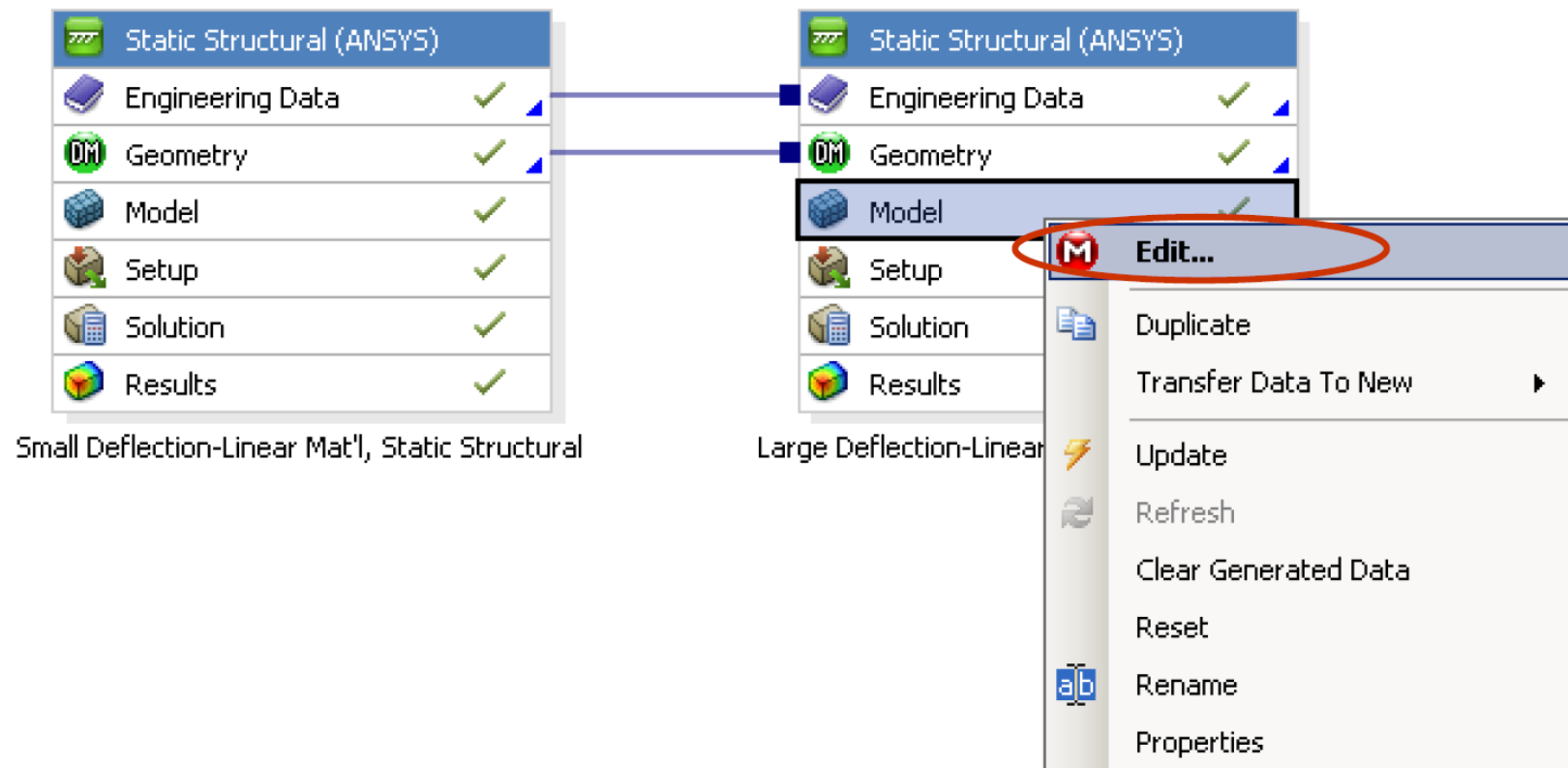
- Highlight the Model Cell, RMB Duplicate...
- Rename this new analysis “Large Deflection - Linear Mat'l, Static Structural”



... Workshop 2A: Large Deflection

Double click on the Large Deflection Model Cell to open the FE Model (Mechanical Session) (or RMB=>Edit...)

Project Schematic



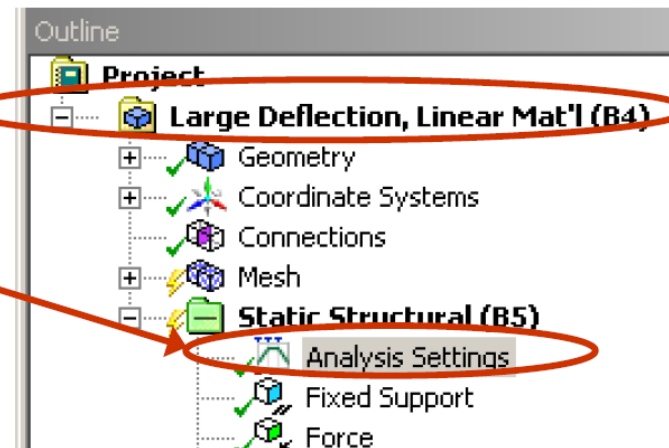
... Workshop 2A: Large Deflection

Highlight the Model folder and change the name to "Large Deflection..."

Highlight the Analysis Settings Folder

- Turn on Large Deflection

Run the new Solution...



Details of "Analysis Settings"

Step Controls

Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled

Solver Controls

Solver Type	Program Controlled
Weak Springs	Off
Large Deflection	On
Inertia Relief	Off

Nonlinear Controls

Output Controls

Analysis Data Management

Visibility

... Workshop 2A: Large Deflection

After solution run is complete, open the Solution Information folder and scroll to near the bottom of the output. Note the solution now solves with 15 cumulative iterations made on the stiffness matrix during the run to account for large deflection effects.

The screenshot shows the ANSYS Workbench interface. On the left, the Project tree is expanded to show the 'Solution (B6)' folder, which includes 'Solution Information', 'Total Deformation', 'Equivalent Stress', and 'Force Reaction'. The 'Solution Information' folder is selected. On the right, the Worksheet displays the output for the first load step. A red circle highlights the convergence results for the first load step, showing that the solution converged after 15 cumulative iterations.

```

Worksheet
  FORCE CONVERGENCE VALUE = 78.28 CRITERION= 1.884
  MOMENT CONVERGENCE VALUE = 0.6259 CRITERION= 6.783 <<< CONVERGED
  DISP CONVERGENCE VALUE = 0.3567E-02 CRITERION= 1.026 <<< CONVERGED
  EQUIL ITER 9 COMPLETED. NEW TRIANG MATRIX. MAX DOF INC= -0.3567E-02
  FORCE CONVERGENCE VALUE = 0.2634E-01 CRITERION= 1.106 <<< CONVERGED
  MOMENT CONVERGENCE VALUE = 0.1226E-01 CRITERION= 6.922 <<< CONVERGED
  >>> SOLUTION CONVERGED AFTER EQUILIBRIUM ITERATION 9

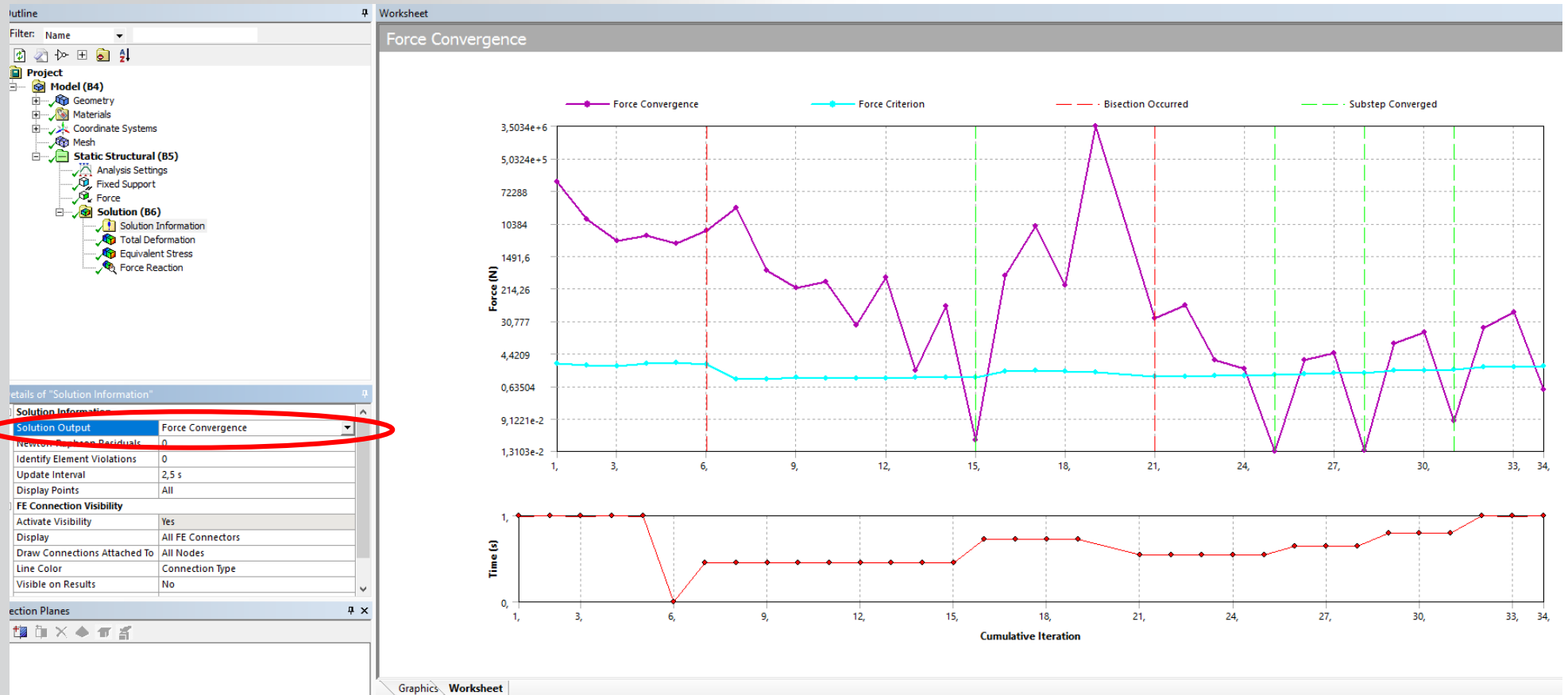
  *** ELEMENT RESULT CALCULATION TIMES
  TYPE NUMBER ENAME TOTAL CP AVE CP
  1 1182 SHELL181 0.109 0.000093
  2 8 SURF156 0.000 0.000000

  *** NODAL LOAD CALCULATION TIMES
  TYPE NUMBER ENAME TOTAL CP AVE CP
  1 1182 SHELL181 0.000 0.000000
  2 8 SURF156 0.000 0.000000
  *** LOAD STEP 1 SUBSTEP 1 COMPLETED. CUM ITER = 15
  *** TIME = 0.450000 TIME INC = 0.450000
  *** AUTO TIME STEP: NEXT TIME INC = 0.27500 DECREASED (FACTOR = 0.6111)

  FORCE CONVERGENCE VALUE = 0.2133E+05 CRITERION= 1.485
  
```

... Workshop 2A: Large Deflection

Change Solution Output to Force Convergence to review the Newton-Raphson History.



... Workshop 2A: Large Deflection

- Review the large deflection stress and displacement results and compare with the first run.
- This is an example of how changing shape and stress stiffening effects can have a subtle influence on the results of what would otherwise be a linear problem.

Please keep this model open for use in Workshop 2B

